MATH 0118

Introduction to Data Science

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Today:

What is Data Science?

What will we be doing in this class?

Tools for Data Science

Installing some tools & Submitting your first assignment!

What is Data Science?

What is Data Science?

- From RStudio CTO and *R for Data Science* co-author Hadley Wickham: “Data science is the process of turning data into understanding.”

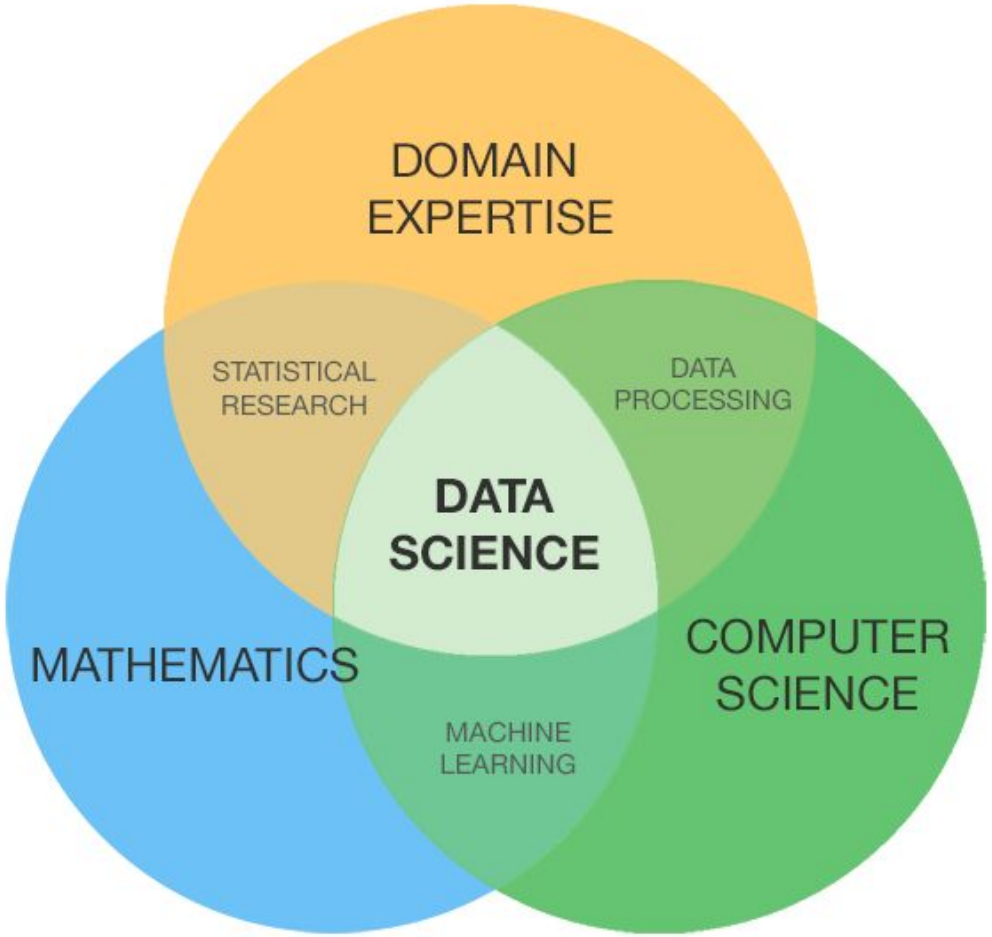
We use data to understand something

- What customers are interested in my product?
- Who will respond to this cancer treatment?
- How do neighborhoods in Chicago differ?

We use data to do something

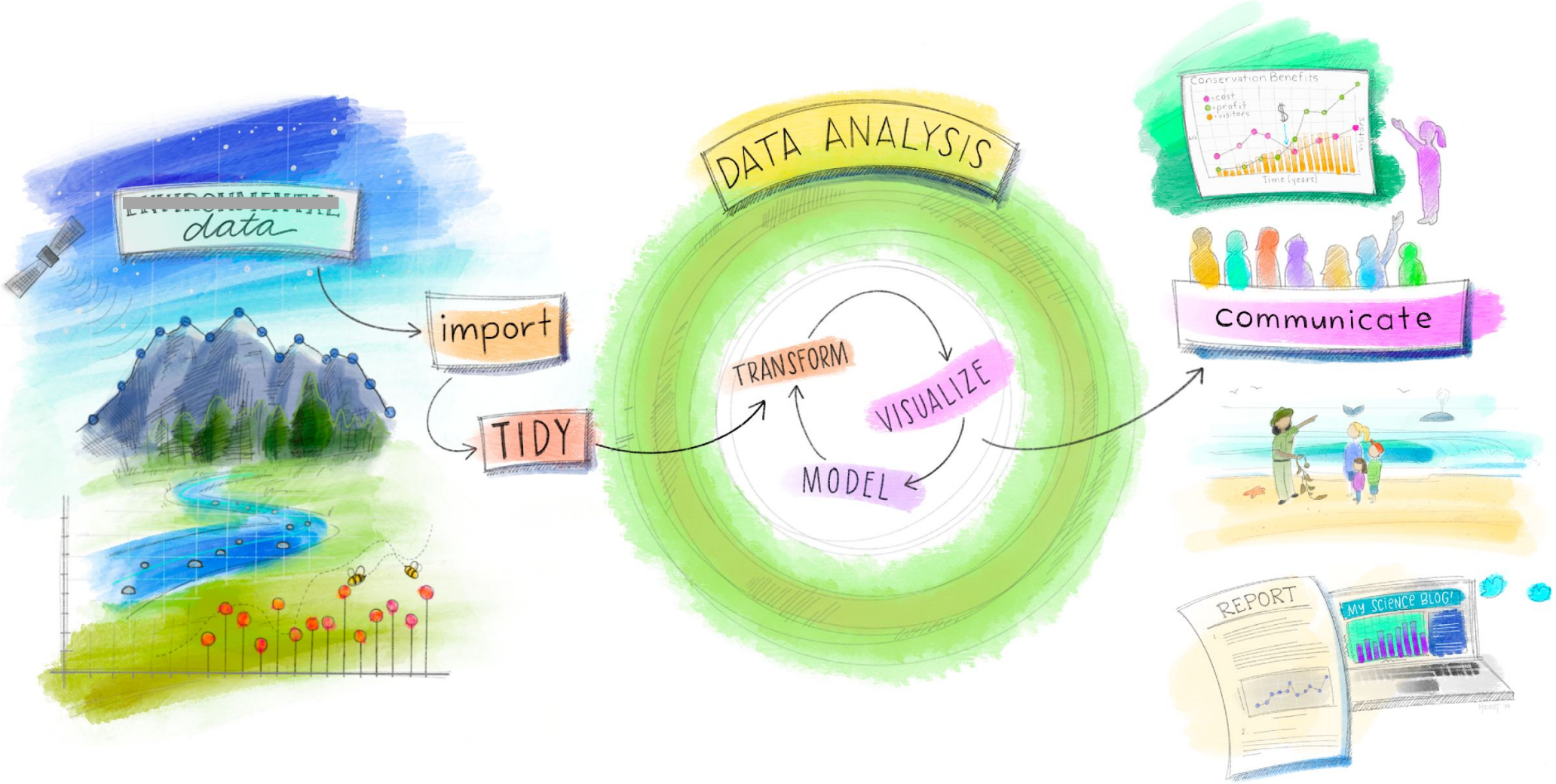
- Facial recognition
- Targeted Advertising
- Predict Disease Spread
- Fraud Detection

Who is a Data Scientist?



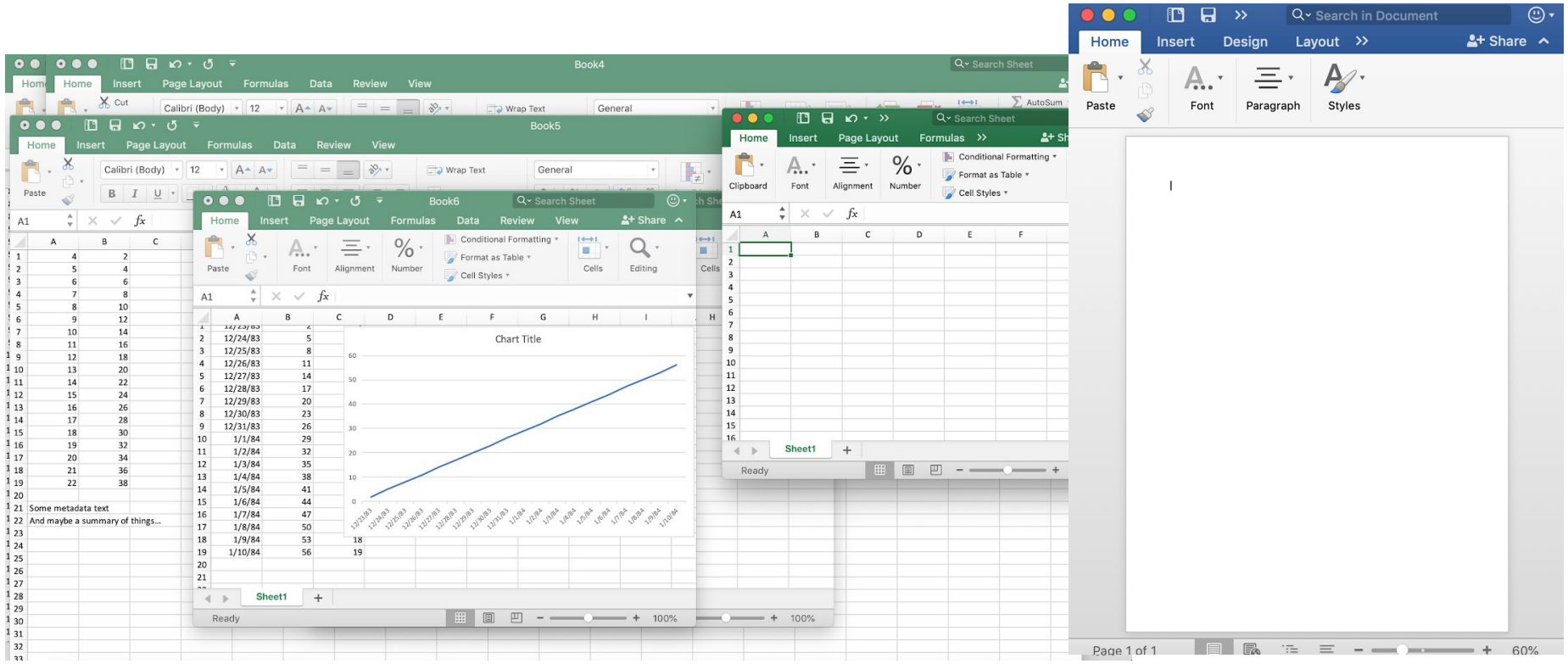
Source: Palmer, Shelly. Data Science for the C-Suite. New York: Digital Living Press, 2015. Print.

Data Science Process



@allisonhorst & @jules32

Is this ↓ how you've been working with data?



Great! But what about this workflow might make it hard to reproduce?

Tools for this Course:

R

- R is a language (predecessor: S language) w/ environment
- Free & open source
- Think “big fancy calculator”

R Studio

- An **I**ntegrated **D**evelopment **E**nvironment for working w/ R - it makes it easier to write R code, work with objects, prepare & publish documents (e.g. bookdown, blogdown, Shiny), and much more all in one place

R Markdown

- R Markdown is a file format for making dynamic documents with R. An R Markdown document is written in markdown (an easy-to-write plain text format) and contains chunks of embedded R code
- Think Word document or Google Doc with code in it

R Packages